



PIRLS Reading Practice — Year 5

Providence English Private School

Set 6: How Volcanoes Are Born

Informational Text · 13 Questions · 40 Minutes

Name: _____ Class: _____ Date: _____

How Volcanoes Are Born

Deep beneath our feet, far below the ground we walk on, the Earth is far from solid and still. In fact, our planet is made up of several layers, and one of these layers — the mantle — is so hot that some of the rock within it slowly melts, becoming a thick, flowing substance called magma. Volcanoes are, in essence, the result of this magma finding its way to the surface.

The Earth's outer layer, called the crust, is not one single, unbroken shell. Instead, it is broken into several enormous pieces called tectonic plates, which fit together like a cracked eggshell and float slowly on the softer layer beneath them. These plates are constantly moving, though so slowly — often just a few centimetres each year — that we rarely notice.

Most volcanoes form along the edges where two tectonic plates meet. In some places, one plate is forced beneath another, sinking into the hot mantle below in a process scientists call subduction. As the sinking plate melts, it creates large pools of magma, which is less dense than the solid rock surrounding it. Because of this, the magma slowly rises upward, searching for cracks or weak points in the crust above.

When magma finally reaches the surface, it is called lava. The way a volcano erupts depends heavily on how thick or runny this magma is. Thin, runny magma allows gases to escape easily, often producing relatively gentle eruptions where lava flows steadily downhill, like the volcanoes found in Hawaii. Thick, sticky magma, on the other hand, traps gases underneath it, building up enormous pressure until it eventually escapes in a sudden, violent explosion — the kind of eruption most people picture when they imagine volcanoes.

Not all volcanoes form at the edges of tectonic plates. Some, known as "hotspot volcanoes," form in the middle of a plate, where an unusually hot area deep within the mantle melts through the crust above it. The Hawaiian Islands were formed this way, created one by one as the Pacific plate slowly moved over a fixed hotspot beneath it, like a piece of paper being dragged slowly over a candle flame.

While volcanic eruptions can be destructive, they also play an important role in shaping our planet. Volcanic ash, once it breaks down, creates extremely fertile soil, which is why many farming communities have historically settled near volcanoes despite the risks involved. Volcanic activity has also released gases over billions of years that helped form Earth's early atmosphere and oceans.

Scientists monitor active volcanoes closely, watching for warning signs such as small earthquakes, changes in gas emissions, or ground swelling caused by rising magma. While predicting the precise moment of an eruption remains extremely difficult, these warning signs can sometimes provide crucial time for nearby communities to evacuate safely.

Volcanoes serve as a dramatic reminder that, despite how solid and unchanging our planet might seem from day to day, powerful forces are constantly at work beneath the surface, slowly and continuously reshaping the world we live on.

Questions

1. MULTIPLE CHOICE · RETRIEVE

What is magma?

- ☐ Rock that has cooled and hardened
- ☐ Melted rock found within the Earth's mantle
- ☐ A type of volcanic ash
- ☐ Gas released during an eruption

2. MULTIPLE CHOICE · RETRIEVE

What are tectonic plates?

- ☐ Large pieces that make up the Earth's crust
- ☐ Types of volcanic rock
- ☐ Layers of magma
- ☐ Tools scientists use to study volcanoes

3. WRITTEN RESPONSE · INTERPRET & INTEGRATE

Explain, in your own words, what 'subduction' means and how it relates to magma formation.

4. MULTIPLE CHOICE · RETRIEVE

What is lava, according to the passage?

- ☐ Magma that has reached the Earth's surface
- ☐ A type of tectonic plate
- ☐ Ash produced by an eruption
- ☐ Cooled volcanic rock

5. WRITTEN RESPONSE · INTERPRET & INTEGRATE

Why do thin, runny magma and thick, sticky magma produce such different types of eruptions?

6. MULTIPLE CHOICE · RETRIEVE

What type of eruption is associated with the volcanoes in Hawaii, according to the passage?

- ☐ Sudden, violent explosions
- ☐ Relatively gentle eruptions with steadily flowing lava
- ☐ No eruptions occur there
- ☐ Only underwater eruptions

7. MULTIPLE CHOICE · RETRIEVE

How were the Hawaiian Islands formed, according to the passage?

- ☐ By a single enormous eruption
- ☐ By a plate moving slowly over a fixed hotspot in the mantle
- ☐ By two plates colliding directly
- ☐ By meteorite impacts

8. WRITTEN RESPONSE · EVALUATE

The passage compares a hotspot volcano to 'a piece of paper being dragged slowly over a candle flame.' What does this comparison help explain?

9. MULTIPLE CHOICE · RETRIEVE

According to the passage, why do some farming communities settle near volcanoes despite the risks?

- (A) Volcanoes provide fresh water
- (B) Volcanic ash creates extremely fertile soil
- (C) Volcanoes attract tourists who buy their crops
- (D) There is no risk involved at all

10. MULTIPLE CHOICE · RETRIEVE

What warning signs do scientists watch for before a volcanic eruption?

- (A) Only changes in temperature
- (B) Small earthquakes, gas emission changes, and ground swelling
- (C) The colour of the sky
- (D) Increased tourism in the area

11. WRITTEN RESPONSE · RETRIEVE

Based on the passage, is it currently possible for scientists to predict the exact moment a volcano will erupt? Explain using evidence from the text.

12. MULTIPLE CHOICE · EVALUATE

What is the main purpose of the final paragraph of the passage?

- (A) To list more types of volcanoes
- (B) To summarise the idea that powerful forces are constantly reshaping the Earth beneath the surface
- (C) To introduce a brand new fact not mentioned before
- (D) To argue that volcanoes should be avoided entirely

13. WRITTEN RESPONSE · EVALUATE

The passage mentions that volcanic eruptions can be both destructive and beneficial. Do you think this makes volcanoes more interesting to study? Explain your thinking, using ideas from the passage.

Answer Key & Model Answers

1. B — Melted rock found within the Earth's mantle

The passage defines magma as melted rock that forms in the hot mantle layer.

2. A — Large pieces that make up the Earth's crust

The passage explains the crust is broken into large pieces called tectonic plates.

3. Written response

Model answer: Subduction is when one tectonic plate is forced underneath another, sinking down into the hot mantle; as the sinking plate melts from the heat, it creates large pools of magma.

Marking guidance: Full credit: explains one plate sinking beneath another AND connects it to magma being created from the melting. Partial credit: mentions plates moving without explaining the melting/magma connection.

4. A — Magma that has reached the Earth's surface

The passage states that when magma reaches the surface, it is called lava.

5. Written response

Model answer: Thin, runny magma lets gases escape easily, causing gentler eruptions with steadily flowing lava, while thick, sticky magma traps gases underneath it, building up pressure that eventually causes a sudden, violent explosion.

Marking guidance: Full credit: explains the gas-escape mechanism for both magma types and links it to eruption type. Partial credit: says one type is 'gentler' and one 'more violent' without explaining the gas-trapping reason.

6. B — Relatively gentle eruptions with steadily flowing lava

The passage describes Hawaiian volcanoes as having gentler eruptions due to thin, runny magma.

7. B — By a plate moving slowly over a fixed hotspot in the mantle

The passage explains the islands formed as the Pacific plate moved over a fixed hotspot beneath it.

8. Written response

Model answer: This comparison helps explain how a moving plate can create a chain of separate volcanic islands over time — just as different spots on a piece of paper would burn one after another as it's dragged over a fixed flame, different parts of the plate get melted through as they pass over the fixed hotspot.

Marking guidance: Full credit: explains that the comparison illustrates how a fixed heat source creates a sequence of effects on something moving above it. Partial credit: restates the comparison without explaining what it clarifies.

9. B — Volcanic ash creates extremely fertile soil

The passage explains that volcanic ash creates fertile soil, which is why farmers settle nearby.

10. B — Small earthquakes, gas emission changes, and ground swelling

The passage lists these three specific warning signs scientists monitor.

11. Written response

Model answer: No — the passage states that predicting the precise moment of an eruption 'remains extremely difficult,' even though warning signs can give communities some time to evacuate.

Marking guidance: Full credit: correctly states it is not currently possible to predict exactly, with supporting quote/detail. Partial credit: says 'no' without referencing the text.

12. B — To summarise the idea that powerful forces are constantly reshaping the Earth beneath the surface

The final paragraph draws together the passage's ideas into a broader reflection on ongoing geological change.

13. Written response

Model answer: Answers will vary, but a thoughtful response might argue that this dual nature makes volcanoes especially fascinating to study, since understanding them helps us both protect communities from danger and appreciate their vital role in shaping fertile soil, the atmosphere, and oceans over Earth's history.

Marking guidance: Full credit: gives a reasoned opinion connecting both the destructive and beneficial aspects mentioned in the passage. Partial credit: mentions only one aspect (destructive or beneficial) in the reasoning.